ENT ATLAS

IMAGING

Imaging conditions: Mouth, throat and salivary glands

Key facts and case images for the core conditions, then the advanced and rare

· November 2026

IMAGING

Core conditions

The common conditions every resident should recognise, each with its key imaging findings, top differentials, a short checklist and representative scans.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Adenoid Hypertrophy

IMAGING FINDINGS

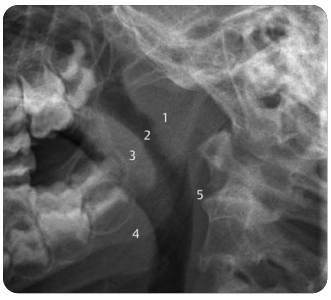
- Symmetric soft tissue in the nasopharyngeal roof and posterior wall
- Striated pattern on T2 with uniform enhancement
- Narrowed nasopharyngeal airway on a lateral film

TOP DIFFERENTIALS

- Nasopharyngeal carcinoma or lymphoma in an adult
- Thornwaldt cyst
- Juvenile angiofibroma

CHECKLIST

- In adults, asymmetric or bulky adenoids need endoscopy
- Middle ear effusions



Plain radiograph for adenoid evaluation.
Dunnebier, Imaging for Otolaryngologists, Fig. 7.2

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

Nasopharyngeal Carcinoma

IMAGING FINDINGS

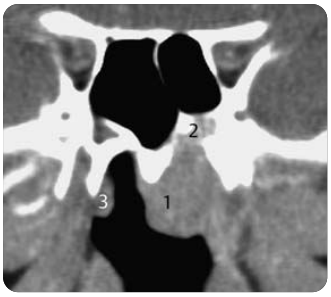
- Asymmetric mass in the fossa of Rosenmüller
- Parapharyngeal and skull base spread; clival marrow replaced on T1
- Retropharyngeal and level II and V nodes

TOP DIFFERENTIALS

- Adenoidal lymphoid hyperplasia (symmetric, striated)
- Lymphoma
- Minor salivary gland tumour

CHECKLIST

- MRI for staging: skull base, cavernous sinus, perineural spread
- Middle ear effusion from Eustachian tube obstruction



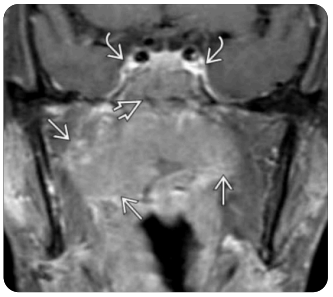
CT, coronal with intravenous contrast enhancement. Note the soft-tissue mass in the roof of the nasopharynx of the left side (1), with invasion in the…



Coronal NECT of paranasal sinuses, performed in a 62-year-old Asian man for evaluation of nasal stuffiness and epistaxis, demonstrates an…



Sagittal T1WI MR in the same patient reveals a large soft tissue mass in the nasopharynx extending superiorly to the sphenoid and sella. There is…



Coronal T1WI C+ FS MR reveals moderate enhancement of a bulky mass filling the nasopharynx with an indistinct right lateral margin.…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 484.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Tonsillar Abscess

IMAGING FINDINGS

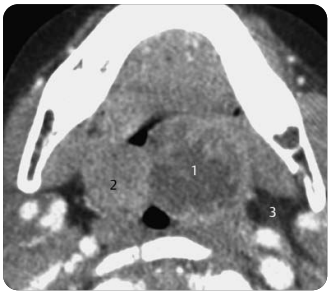
- Rim-enhancing fluid collection in or around the tonsil
- Swollen tonsil and pharyngeal wall
- Spread to parapharyngeal, masticator or carotid space

TOP DIFFERENTIALS

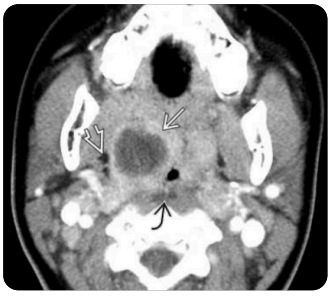
- Tonsillitis without abscess (phlegmon)
- Tonsillar retention cyst
- Retropharyngeal abscess; tonsillar carcinoma

CHECKLIST

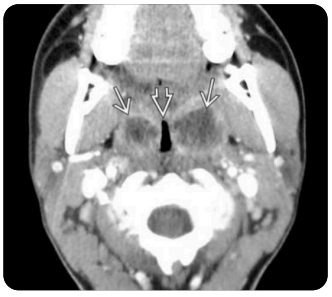
- Airway
- Internal jugular vein thrombosis and carotid



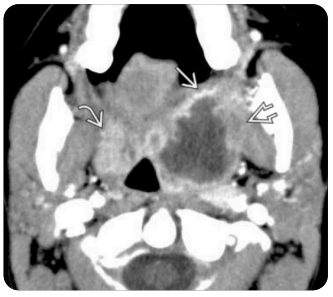
CT with intravenous contrast enhancement, axial. The left tonsil (1) is much larger than the contralateral side (2) with central hypodensities…



Axial CECT shows a large, low-density tonsillar abscess. No extension through the capsule into PPS is present. The left tonsil is prominent…



Axial CECT reveals bilateral tonsillar abscesses, the left larger than the right. Note the mass effect on both PPS but no extension of inflammatory…



Axial CECT demonstrates left tonsillar and peritonsillar abscess with extension through the capsule into the posterior buccal space and medial…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 28.

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

Tonsillar Carcinoma

IMAGING FINDINGS

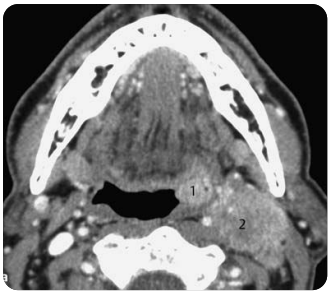
- Asymmetric enhancing tonsillar mass, sometimes small
- Spread to soft palate, tongue base and parapharyngeal space
- Cystic or necrotic level II nodes (HPV)

TOP DIFFERENTIALS

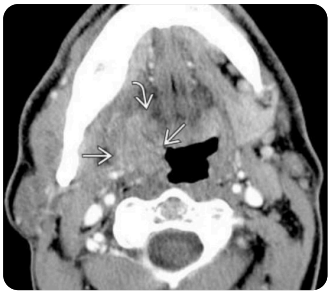
- Tonsillar lymphoid hyperplasia
- Lymphoma
- Peritonsillar abscess

CHECKLIST

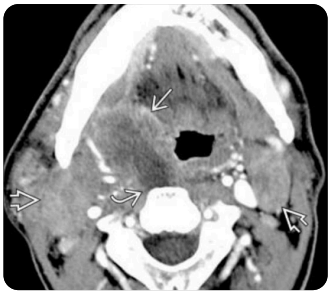
- Medial pterygoid, mandible and carotid
- Both sides of the neck



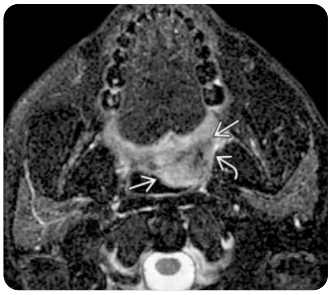
CT with intravenous contrast enhancement, axial. In the left tonsillar fossa, a small and enhancing tumor is visible (1), with enlarged level II lymph…



Axial CECT in alcoholic smoker presenting with multiple right neck masses shows an enlarged right palatine tonsil mass…



Axial CECT, same patient 1 year later, shows dramatic progression of disease. Stage is now T4a N2c M1. The tumor is much larger, extending into…



Axial T2WI FS MR in a 50-year-old presenting with a palatine tonsil mass shows heterogeneous exophytic left tonsillar tumor. Well-defined lateral…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 492.

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

Oropharyngeal Tumor, Base of Tongue

IMAGING FINDINGS

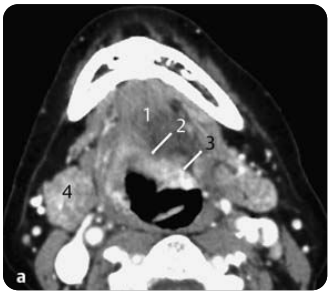
- Asymmetric fullness of the lingual tonsil with enhancement
- Spread to vallecula, pre-epiglottic space and floor of mouth
- Level II nodes, often bilateral

TOP DIFFERENTIALS

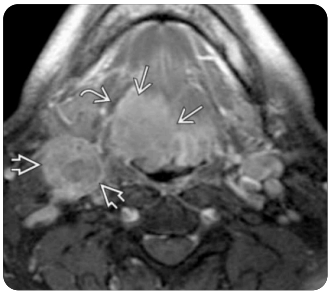
- Lingual tonsil lymphoid hyperplasia
- Lymphoma
- Thyroglossal or vallecular cyst

CHECKLIST

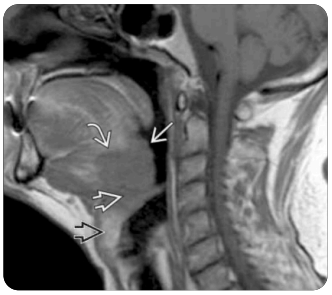
- Crossing the midline and neurovascular bundles
- Pre-epiglottic and laryngeal invasion



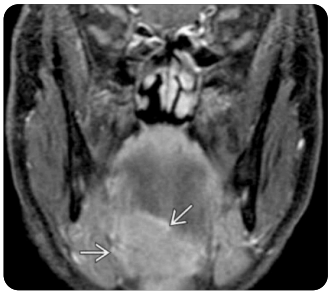
CT with intravenous contrast enhancement, axial. Axial slice at the level of the mandible and floor of the mouth (1). At clinical evaluation, an…



Axial T1WI C+ FS MR in a 58-year-old alcoholic presenting with persistent right neck mass after dental procedure demonstrates enlarged,…



Sagittal T1WI MR in the same patient shows fullness at the right base of tongue with a tumor replacing the lingual tonsillar tissue and infiltrating…



Coronal T1WI C+ FS MR shows an asymmetric mass at the right base of tongue in the same patient. This was found to be an invasive SCCa and…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 488.

For personal study. Follow local protocols and senior advice for patients.

SALIVARY GLANDS · INFECTION & INFLAMMATION

Sialadenitis

IMAGING FINDINGS

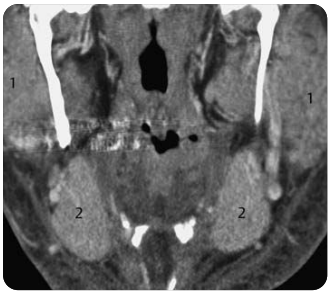
- Enlarged, enhancing gland with stranding of the surrounding fat
- Dilated duct, often with a stone
- Abscess: rim-enhancing collection

TOP DIFFERENTIALS

- Dental infection
- Nodal metastasis
- Gland tumour

CHECKLIST

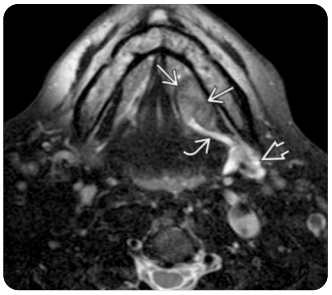
- Look for a stone along the duct
- Chronic: small, fibrotic gland



Patient with chronic intermittent enlargement of the salivary glands and diffuse pain.
Dunnebier, Imaging for…



Axial CECT demonstrates an asymmetrically enhancing, enlarged left SMG compared to the right. Calculus is evident in the distal…



Axial T2 FS MR demonstrates a well-circumscribed mass in the left floor of mouth in a patient with mucosal SCCa. The mass is obstructing the left…



Axial CECT through floor of mouth in a patient with right submandibular "mass" shows dense calculus in proximal duct at SMG hilus. Note…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 406.

SALIVARY GLANDS · INFECTION & INFLAMMATION

Sialolith of the Submandibular Gland

IMAGING FINDINGS

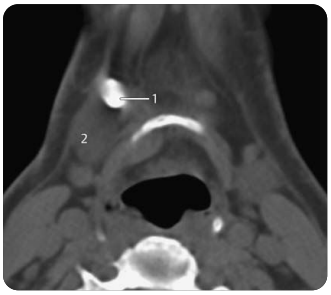
- Dense calculus in the Wharton duct or hilum on non-contrast CT
- Dilated duct proximal to the stone
- Enlarged inflamed or atrophic gland

TOP DIFFERENTIALS

- Phlebolith (venous malformation)
- Calcified node
- Dental fragment

CHECKLIST

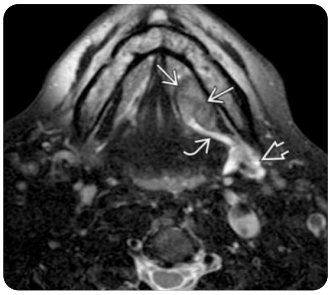
- Position relative to the mylohyoid and hilum
- Ultrasound shows stone and duct without radiation



CT, axial bone window.A hyperdensity (1), with similar density as bone, suggestive of a sialolith (stone, calculus), probably obstructin outflo…



Axial CECT demonstrates an asymmetrically enhancing, enlarged left SMG compared to the right. Calculus is evident in the distal…



Axial T2 FS MR demonstrates a well-circumscribed mass in the left floor of mouth in a patient with mucosal SCCa. The mass is obstructing the left…



Axial CECT through floor of mouth in a patient with right submandibular "mass" shows dense calculus in proximal duct at SMG hilus. Note…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 406.

SALIVARY GLANDS · INFECTION & INFLAMMATION

Ranula of the Sublingual Gland

IMAGING FINDINGS

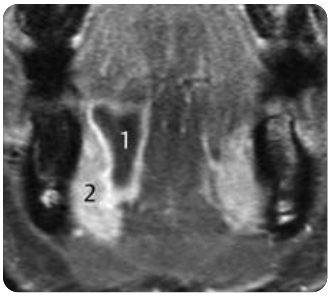
- Simple ranula: thin-walled cyst in the sublingual space
- Diving ranula: passes behind or through the mylohyoid into the submandibular space
- Tail sign pointing back to the sublingual space

TOP DIFFERENTIALS

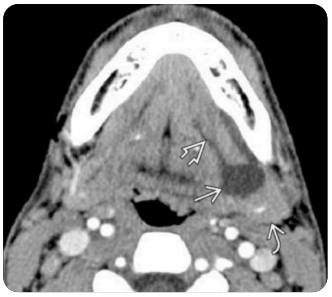
- Lymphatic malformation
- Dermoid or epidermoid
- Second branchial or thyroglossal cyst

CHECKLIST

- Relation to the mylohyoid decides the approach
- Remove the sublingual gland to prevent recurrence



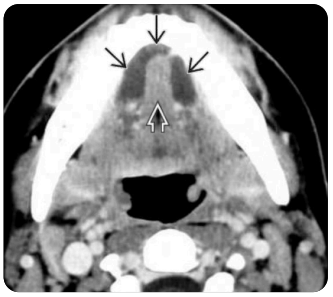
MRI, T1-weighted with gadolinium enhancement and fat suppression, coronal. There is a cystic lesion suggestive of a ranula (1), originating…



Axial CECT demonstrates a rounded cystic mass in the submandibular space abutting the posterior margin of the mylohyoid muscle and displacing…



Axial CECT shows simple unilateral ranula as an ovoid, low-density lesion in left anterior SLS. There is no perceptible thickening or…



Axial CECT demonstrates a cystic, horseshoe-shaped lesion in anterior SLS, curving around and above genioglossus muscle insertion on…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 400.

SALIVARY GLANDS · TUMOURS

Pleomorphic Adenoma of the Parotid Gland

IMAGING FINDINGS

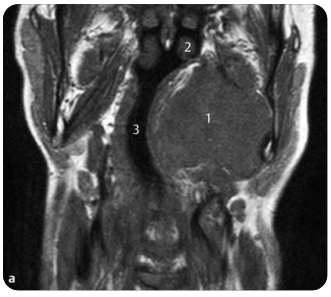
- Well-defined, lobulated mass with a capsule
- Very bright on T2
- Deep lobe: widens the stylomandibular gap, pushes parapharyngeal fat medially

TOP DIFFERENTIALS

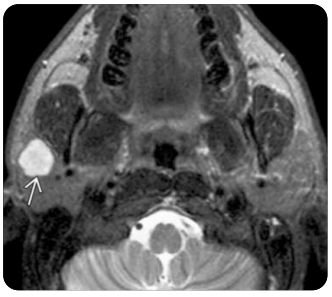
- Warthin tumour (tail, multiple, smokers)
- Low-grade mucoepidermoid carcinoma
- Intraparotid node

CHECKLIST

- Superficial or deep to the retromandibular vein
- Never enucleate: recurrence



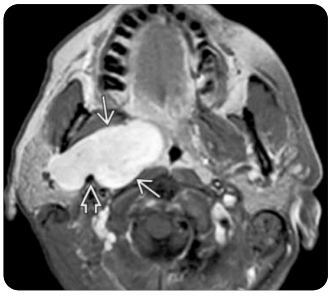
MRI, T1-weighted, coronal. Note the extensive, isointense, smooth-bordered lesion (1) in the hypopharyngeal area: from the level o…



Axial T2 FS MR reveals a sharply circumscribed, high-signal BMT in the superficial lobe of the parotid gland. This tumor at the very least abuts, if…



Axial T2 FS MR shows a pear-shaped mass extending from the deep lobe of the parotid gland, through the enlarged stylomandibular notch,…



Axial T1WI C+ FS MR shows a large, lobulated, homogeneously enhancing mass in the deep lobe of the parotid. The styloid process indents the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 88.

For personal study. Follow local protocols and senior advice for patients.

SALIVARY GLANDS · TUMOURS

Malignancies of the Parotid Gland

IMAGING FINDINGS

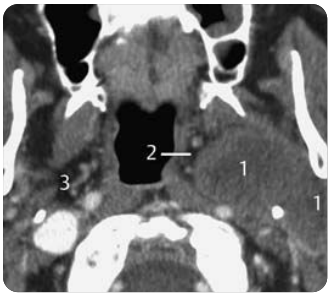
- Ill-defined or infiltrative margins
- Low or intermediate T2, low ADC
- Facial nerve: enhancement through the stylomastoid foramen

TOP DIFFERENTIALS

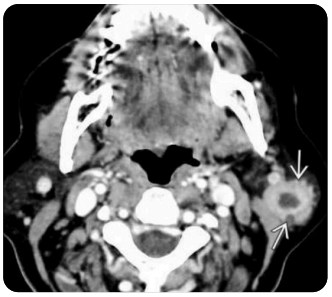
- Pleomorphic adenoma
- Warthin tumour
- Metastatic node from skin cancer; lymphoma

CHECKLIST

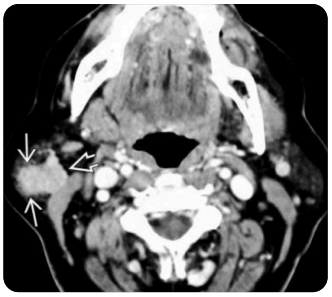
- Perineural spread to the mastoid segment and beyond
- Nodes in the parotid and level II



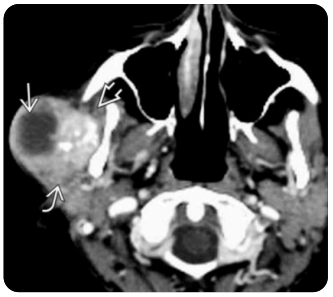
Slight swelling of the left parotid region and deep pain.
Dunnebier, Imaging for Otolaryngologists, Fig. 9.30 a



Axial CECT reveals a well-defined mass in the superficial lobe of the parotid gland. It has a thick rind of peripheral enhancement and is centrally cystic …



Axial CECT shows a mass in the parotid gland with ill-defined lateral borders and well-defined medial borders. The tumor enhances uniformly. Although…



Axial CECT shows a large, heterogeneous mass with cystic areas, heterogeneous enhancement, and ill-defined borders. The masseter…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 98.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Zenker Diverticulum (Pharyngeal Pouch)

IMAGING FINDINGS

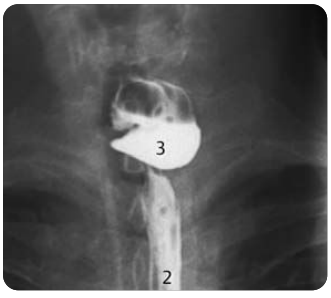
- Outpouching from the posterior hypopharynx above the cricopharyngeus on barium swallow
- Air–fluid level behind the oesophagus on CT
- Residual barium after the swallow

TOP DIFFERENTIALS

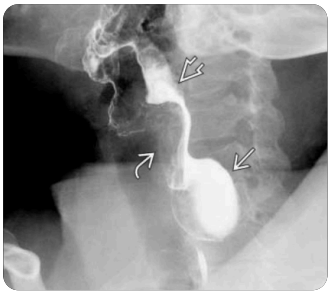
- Lateral pharyngeal or Killian–Jamieson diverticulum
- Paratracheal air cyst

CHECKLIST

- Size and neck of the pouch
- Aspiration on the swallow



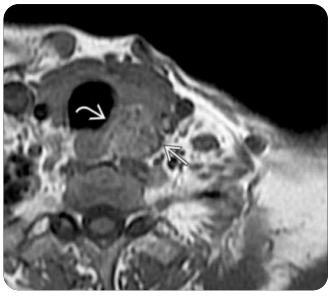
Barium swallow, anteroposterior view. After swallowing contrast, some contrast has remained in the hypopharynx (1) due to physiologic…



Barium esophagram demonstrates a large diverticulum with retained layering contrast from the posterior lateral junction of the hypopharynx…



Parasagittal T2WI MR shows a saccular well-circumscribed collection of heterogenous signal just below the cricoid cartilage at the level of C5-C6.…



Axial T1WI MR of brachial plexus in the same patient shows an incidental finding of a left Zenker diverticulum. Image demonstrates the mouth of th…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 258.

PHARYNX, TONSILS & OESOPHAGUS · TRAUMA

Foreign Bodies

IMAGING FINDINGS

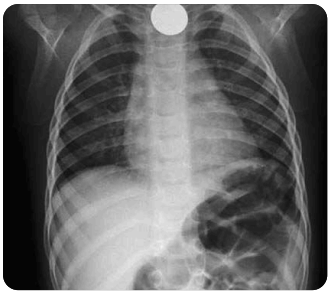
- Radio-opaque body on the lateral neck film; fish bones often faint
- CT shows the object, its position and depth
- Prevertebral swelling, air or abscess after perforation

TOP DIFFERENTIALS

- Calcified cartilage (thyroid, cricoid, arytenoid)
- Calcified stylohyoid ligament

CHECKLIST

- Low-dose CT if the film is negative and suspicion is high
- Perforation: mediastinal air, collection



Conventional radiography can still be convenient for evaluation of radiopaque foreign bodies, as in this 2-year-old child, who was suspected of…

ORAL CAVITY & TONGUE · AFTER SURGERY

Squamous cell carcinoma: oral cavity, oropharynx and larynx

IMAGING FINDINGS

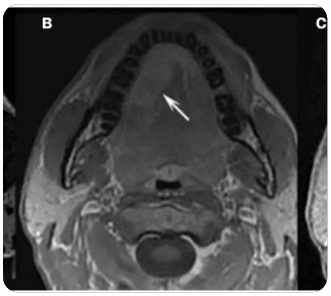
- Enhancing mucosal mass; depth of invasion in the tongue
- Crossing the lingual septum; neurovascular bundles
- Mandible cortex (CT) and marrow (MRI)

TOP DIFFERENTIALS

- Lingual tonsil hyperplasia
- Abscess
- Minor salivary gland tumour

CHECKLIST

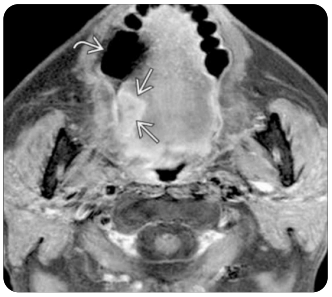
- MRI for oral and oropharyngeal tumours
- Nodes on both sides; chest CT



Axial MRI of an oral cavity SCC shows a solid mass involving the left lateral border of the tongue showing intermediate signal intensity on T2…



Coronal T1WI MR in a woman with a painful right tongue ulcer shows subtle low signal intensity at the lateral tongue margin. There is no evidence …



Axial T1WI C+ FS MR in the same patient shows marked enhancement a of wedge-shaped ulcer at posterolateral tongue surface. No…



Axial T2WI FS MR demonstrates a small right lateral tongue tumor, subtly evident as lenticular region of increased signal intensity. Ill-defined…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 504.

IMAGING

Advanced and rare

Once the core is secure. Each has its full set of images in the app.

Thornwaldt Cyst

Pharynx, tonsils & oesophagus · Congenital

Multiple Synchronous Malignancies

Pharynx, tonsils & oesophagus · Tumours

Staging is based on enlarged or pathologic-appearing lymph nodes, bilaterality, and spread of each tumor to the contralateral side. · Lymphomas (non-Hodgkin) initially may present as unilateral lymphadenopathy and ulcerative lesions.

Pleomorphic Adenoma of the Submandibular Gland

Salivary glands · Tumours

Calcifications are well depicted on CT, as are signs of infection and abscess formation. On CT, most pleomorphic adenomas are smooth-bordered, spherical, benign-looking tumors that usually display a higher density than the surrounding parotid parenchyma. Sometimes, these tumors demonstrate a heterogeneous appearance with sites of lower attenuation representing areas of necrosis, old hemorrhage, and/or cystic change (see · For MRI characteristics, see Pleomorphic Adenoma of the Parotid Gland, page 316.

Lipoma Near the Parotid Gland

Salivary glands · Tumours

Lipomas have fat density on CT (−50 to −100 HU) and follow fat signal on every MRI sequence, with suppression on fat-saturated images. · Thin septa are acceptable; thick enhancing septa or nodules suggest liposarcoma.

Cholesteatoma Near the Parotid Gland

Salivary glands · Infection & inflammation

A cholesteatoma or epidermoid can extend from the ear canal or mastoid into the parotid region, or arise from a first branchial cleft remnant. · CT: a smooth, expansile low-density lesion with bone erosion of the canal or tympanic plate; MRI: restricted diffusion on DWI, the key feature.

Adenoid Cystic Carcinoma

Salivary glands · Tumours

Slow-growing but relentless, with perineural spread along VII (to the stylomastoid foramen and geniculate ganglion) and V3 (foramen ovale) as its hallmark; skip lesions are common. · MRI with fat-saturated post-contrast T1 shows nerve thickening and enhancement and widening of the foramina; image the whole nerve to the brainstem.

Acinic Cell Carcinoma

Salivary glands · Tumours

A low-grade malignancy, mostly in the parotid, often well circumscribed and so easily mistaken for a benign tumour on imaging. · May be cystic or multifocal; up to a few per cent are bilateral.

Esophageal Passage Disorders

Pharynx, tonsils & oesophagus · Infection & inflammation

Patients with removable dentures may lack oral sensory input regarding the size of their food. This may also happen with sharp foreign bodies, which may cause perforations of the esophagus. · Anatomic structures such as the aortic arch or an aberrant right esophageal artery are demonstrated as persistent impressions without stasis of contrast.